Assignment 1 - Git & GitHub Fundamentals

Session 1 - Group B | Individual Lab Report

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Repository: https://github.com/IsmailFF5/git-session1-IsmailFF5

1. Repository Setup

I initialized the local project using **git init** and named its folder **git-session1-IsmailFF5**. My **.gitignore** uses ***.env** to ignore environment files (including sample.env and local.env), ***.log** to ignore ordinary logs, and **!important.log** to make an exception for important.log. I checked the rules with **git check-ignore -v** and inspected the untracked files using **git status**.

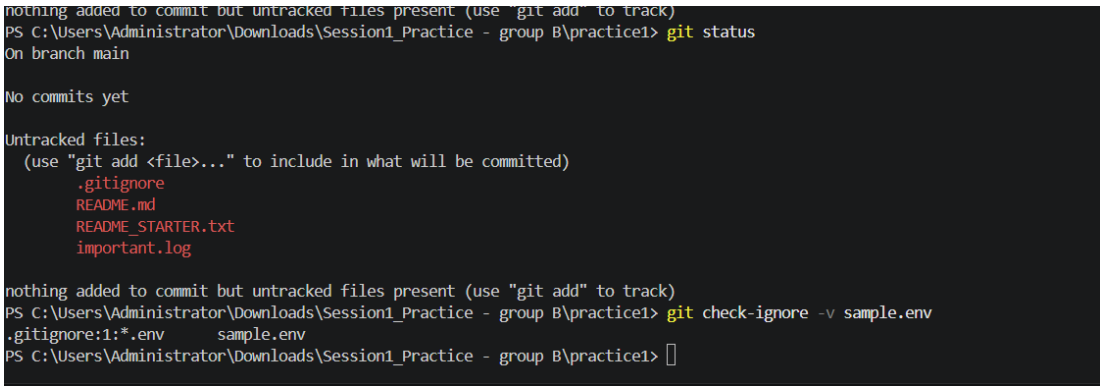


*Figure 1. Git status excludes sample.env; check-ignore confirms the *.env rule.*

2. README & Commit History

My README includes the course title, student details, project description, technologies, a Git command example and a documentation link. I used the starter HTML/CSS files and organized the work into logical commits: initial setup, adding index.html, adding style.css, updating HTML content, and styling the new paragraph. Later commits added About documentation, merged its branch and synchronized a README edit from GitHub. The supplied README image was not available in the starter files examined and remains to be verified.

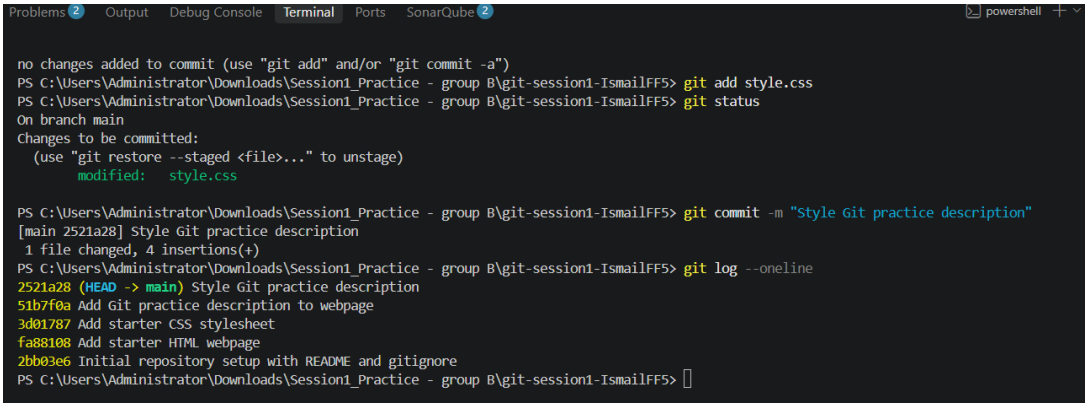


Figure 2. Five meaningful local commits across multiple project files.

3. Selective Staging

I edited both **index.html** and **style.css**. I staged only index.html with **git add index.html**, inspected staged changes using **git diff --staged** and unstaged changes using **git diff**, then committed the HTML update. The next **git status** still showed style.css as modified but not staged. I committed the CSS change separately afterward.

```
+    font-size: 16px;
 }
\ No newline at end of file
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git commit -m "Add Git practi
e description to webpage"
[main 51b7f0a] Add Git practice description to webpage
 1 file changed, 1 insertion(+)
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   style.css

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5>
```

Figure 3. After the HTML commit, style.css remained modified and unstaged.

4. Branch & Merge

I created and switched to **about-feature** with **git switch -c about-feature**. On that branch I added and committed **ABOUT.md**. I returned to **main** using **git switch main** and merged with **git merge --no-ff about-feature**. The explicit merge commit preserved the feature branch relationship in the graph.

```
Problems 2  Output  Debug Console  Terminal  Ports  SonarQube 2                                                    powershell

2521a28 (main) Style Git practice description
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git switch main
Switched to branch 'main'
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git merge --no-ff about-feature -m "Merge About feature into main"
Merge made by the 'ort' strategy.
 ABOUT.md | 12 ++++++++++++
 1 file changed, 12 insertions(+)
 create mode 100644 ABOUT.md
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git log --oneline --graph --all --decorate
*   731da74 (HEAD -> main) Merge About feature into main
|\
| * 52fea32 (about-feature) Add About feature documentation
|/
* 2521a28 Style Git practice description
* 51b7f0a Add Git practice description to webpage
* 3d01787 Add starter CSS stylesheet
* fa88108 Add starter HTML webpage
* 2bb03e6 Initial repository setup with README and gitignore
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5>
```

Figure 4. The commit graph shows the About feature commit and merge into main.

5. Remote Changes

Commit vs push: A commit records staged changes in the local repository; push publishes local commits to a remote repository. I linked GitHub using **git remote add origin** and published main using **git push -u origin main**. **Fetch vs pull:** Fetch updates remote-tracking information without changing the local working files; pull fetches and integrates remote changes. After editing README.md on GitHub, **git fetch origin** showed that local main was one commit behind origin/main. I then used **git pull origin main**, which fast-forwarded main, and verified that the working tree was clean and synchronized.

```
branch 'main' set up to track 'origin/main'.
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.09 KiB | 111.00 KiB/s, done.
From https://github.com/IsmailFF5/git-session1-IsmailFF5
   731da74..12d3080  main       -> origin/main
PS C:\Users\Administrator\Downloads\Session1_Practice - group B\git-session1-IsmailFF5> git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)
```

Figure 5. After fetch, local main was one commit behind origin/main.

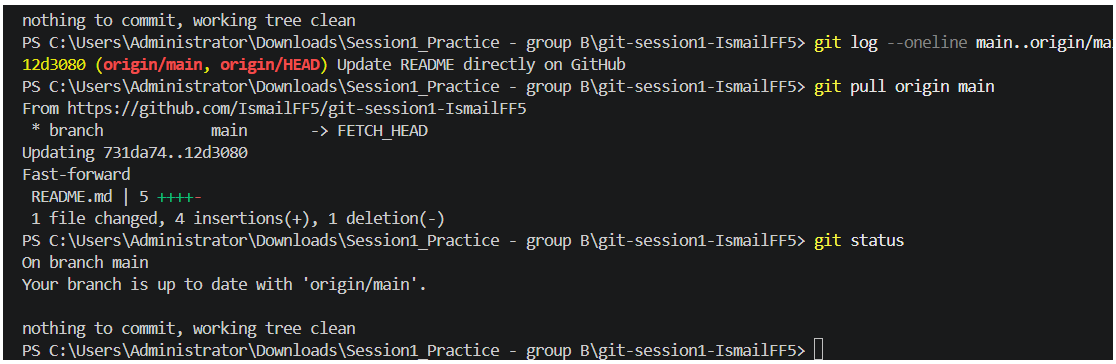


Figure 6. Pull fast-forwarded README.md and restored synchronization.

6. Fair Use of AI

I used ChatGPT as a step-by-step learning assistant while performing the assignment in Visual Studio Code. I asked how to configure .gitignore, interpret git status and check-ignore output, create logical commits, demonstrate selective staging, merge a feature branch, and distinguish fetch from pull. I accepted commands after reviewing their explanations, adjusted the environment ignore rule from .env to *.env to match the provided sample.env file, and checked results against my own terminal output and GitHub repository. I used the actual screenshots from my work as evidence rather than presenting hypothetical results.

7. Sources & Reflection

Sources: Course Assignment 1 and Report Template (Session 1, Group B); Git documentation: https://git-scm.com/docs; my VS Code terminal output and GitHub repository history.

Reflection: The hardest concept was selective staging because two files can be modified while only one is included in a commit. If Git behaves unexpectedly, I would first run **git status** to check the branch and staged/unstaged changes, then inspect **git diff**, **git log** or **git remote -v** as appropriate.

Submission check: The completed PDF must still be added, committed and pushed to the public GitHub repository, and its URL entered in the shared submission sheet. The README image requirement also remains outstanding.